

Linear Functions - Matrix-Vector Multiplication

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1 Matrix-Vector Multiplication

Let $A \in \mathbb{R}^{m \times n}$ be a matrix, $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ be vectors, and $c \in \mathbb{R}$, a scalar.

Matrix-vector multiplication is linear because which satisfies

1. **Additivity:**

$$A(\mathbf{u} + \mathbf{v}) = A\mathbf{u} + A\mathbf{v}.$$

2. **Homogeneity:**

$$A(c\mathbf{u}) = c(A\mathbf{u}).$$

Both properties follow directly from the definition of matrix multiplication and the distributive and associative properties of real numbers. Hence, the function $\mathbf{x} \mapsto A\mathbf{x}$ is linear.

Let us assume representative values for A , $\mathbf{u}$, and $\mathbf{v}$:

$$A = \begin{bmatrix} w_{11} & w_{12} & w_{13} \\ w_{21} & w_{22} & w_{23} \\ w_{31} & w_{32} & w_{33} \\ w_{41} & w_{42} & w_{43} \end{bmatrix}, \quad \mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}, \quad \text{and} \quad \mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$$

1.1 Additivity

Compute $A(\mathbf{u} + \mathbf{v})$:

$$\begin{bmatrix} w_{11} & w_{12} & w_{13} \\ w_{21} & w_{22} & w_{23} \\ w_{31} & w_{32} & w_{33} \\ w_{41} & w_{42} & w_{43} \end{bmatrix} \begin{bmatrix} (u_1 + v_1) \\ (u_2 + v_2) \\ (u_3 + v_3) \end{bmatrix} = \begin{bmatrix} w_{11}(u_1 + v_1) + w_{12}(u_2 + v_2) + w_{13}(u_3 + v_3) \\ w_{21}(u_1 + v_1) + w_{22}(u_2 + v_2) + w_{23}(u_3 + v_3) \\ w_{31}(u_1 + v_1) + w_{32}(u_2 + v_2) + w_{33}(u_3 + v_3) \\ w_{41}(u_1 + v_1) + w_{42}(u_2 + v_2) + w_{43}(u_3 + v_3) \end{bmatrix}$$

Compute $A\mathbf{u}$:

$$\begin{bmatrix} w_{11} & w_{12} & w_{13} \\ w_{21} & w_{22} & w_{23} \\ w_{31} & w_{32} & w_{33} \\ w_{41} & w_{42} & w_{43} \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = \begin{bmatrix} w_{11}u_1 + w_{12}u_2 + w_{13}u_3 \\ w_{21}u_1 + w_{22}u_2 + w_{23}u_3 \\ w_{31}u_1 + w_{32}u_2 + w_{33}u_3 \\ w_{41}u_1 + w_{42}u_2 + w_{43}u_3 \end{bmatrix}$$

Compute $\mathbf{A}\mathbf{v}$:

$$\begin{bmatrix} w_{11} & w_{12} & w_{13} \\ w_{21} & w_{22} & w_{23} \\ w_{31} & w_{32} & w_{33} \\ w_{41} & w_{42} & w_{43} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} w_{11}v_1 + w_{12}v_2 + w_{13}v_3 \\ w_{21}v_1 + w_{22}v_2 + w_{23}v_3 \\ w_{31}v_1 + w_{32}v_2 + w_{33}v_3 \\ w_{41}v_1 + w_{42}v_2 + w_{43}v_3 \end{bmatrix}$$

Compute $\mathbf{A}\mathbf{u} + \mathbf{A}\mathbf{v}$:

$$\begin{bmatrix} w_{11}u_1 + w_{12}u_2 + w_{13}u_3 \\ w_{21}u_1 + w_{22}u_2 + w_{23}u_3 \\ w_{31}u_1 + w_{32}u_2 + w_{33}u_3 \\ w_{41}u_1 + w_{42}u_2 + w_{43}u_3 \end{bmatrix} + \begin{bmatrix} w_{11}v_1 + w_{12}v_2 + w_{13}v_3 \\ w_{21}v_1 + w_{22}v_2 + w_{23}v_3 \\ w_{31}v_1 + w_{32}v_2 + w_{33}v_3 \\ w_{41}v_1 + w_{42}v_2 + w_{43}v_3 \end{bmatrix} = \begin{bmatrix} w_{11}(u_1 + v_1) + w_{12}(u_2 + v_2) + w_{13}(u_3 + v_3) \\ w_{21}(u_1 + v_1) + w_{22}(u_2 + v_2) + w_{23}(u_3 + v_3) \\ w_{31}(u_1 + v_1) + w_{32}(u_2 + v_2) + w_{33}(u_3 + v_3) \\ w_{41}(u_1 + v_1) + w_{42}(u_2 + v_2) + w_{43}(u_3 + v_3) \end{bmatrix}$$

Note that $\mathbf{A}(\mathbf{u} + \mathbf{v}) = \mathbf{A}\mathbf{u} + \mathbf{A}\mathbf{v}$. Thus **Additivity** holds for general matrix-vector multiplication.

1.2 Homogeneity

Prove Homogeneity!